

# Does M&A create value?

786 significant acquisitions by US listed companies, followed for three to five years after closing: shareholder returns against comparable stocks, risk-adjusted alphas, margins and goodwill. Rebuilt from SEC filings, prices including delisted stocks, and Fama-French benchmarks.

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−12.4%

mean 36-month abnormal return vs size and book-to-market peers

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37%

of acquirers beat their benchmark over three years

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−6.8%

yearly five-factor alpha when the deal was paid partly or mostly in stock

# Only deals that matter to the acquirer.

SAMPLE

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Acquirer-years with \$100m+ spent on acquisitions, 2011 to 2022 (XBRL) | 7,304      |
| Outside finance and utilities                                          | 5,596      |
| Completion filed on 8-K Item 2.01, 2012 to 2021                        | 2,007      |
| Matched to a validated price series                                    | 1,666      |
| <b>At least 10% of acquirer value, one deal per 3 years</b>            | <b>786</b> |

Median deal \$618m, 27% of the acquirer's market value. Each ticker match is checked against the public float on the 10-K cover. Firms that delist earn the benchmark for the rest of the window.

TWO WAYS TO MEASURE

## Event time: buy-and-hold

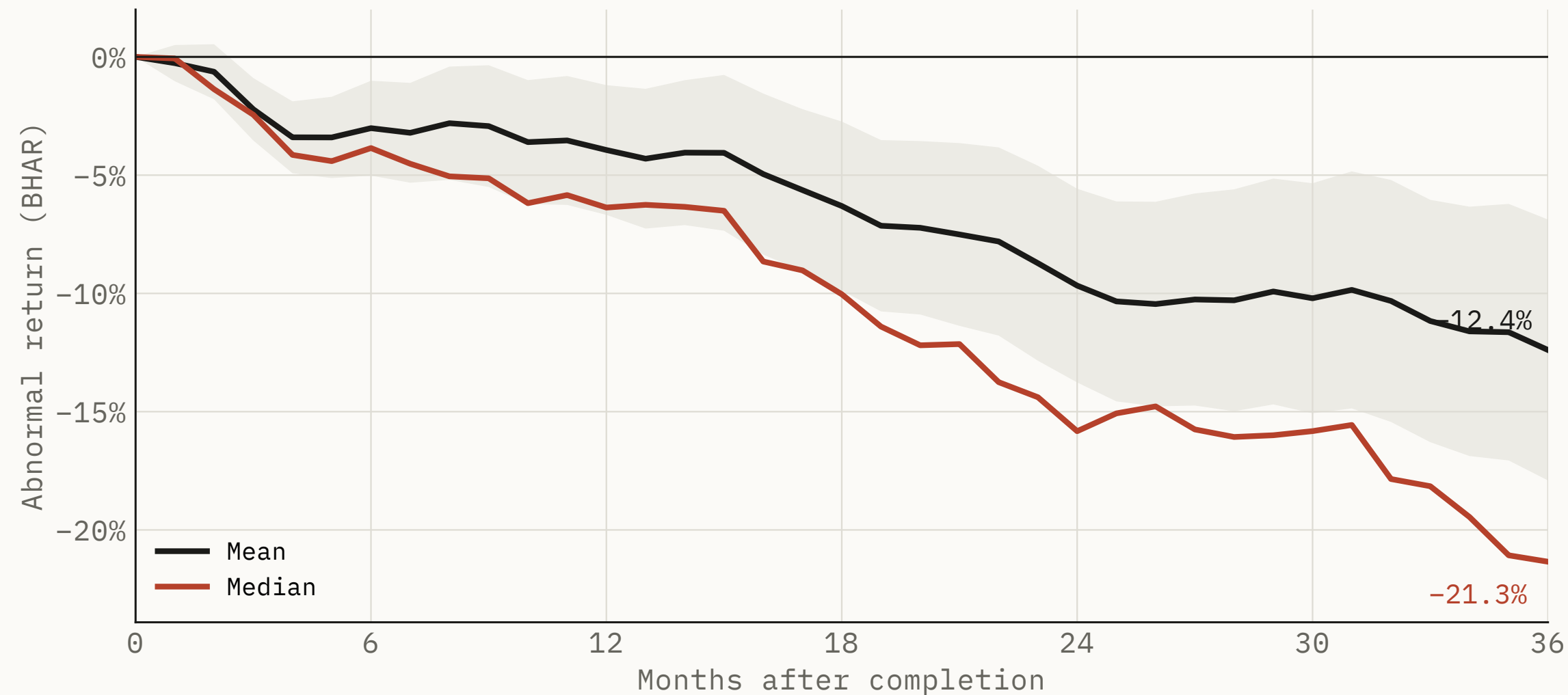
Compound the acquirer's monthly total return from month 1 to 36 and subtract its Fama-French size and book-to-market portfolio. What a shareholder actually earned; skewness-adjusted and bootstrapped tests.

## Calendar time: alpha

Each month, hold every acquirer within 36 months of its deal; regress on five factors plus momentum. Handles overlapping deals and risk exposures; the test Fama (1998) prefers.

# Acquirers fell 12.4% behind their peers in three years.

MEAN AND MEDIAN BHAR, MONTHS AFTER COMPLETION · SHADED: 95% INTERVAL OF THE MEAN



−3.9%

after 12 months

−12.4%

after 36 months · median −21.3% ·  $t = -4.11$

−18.6%

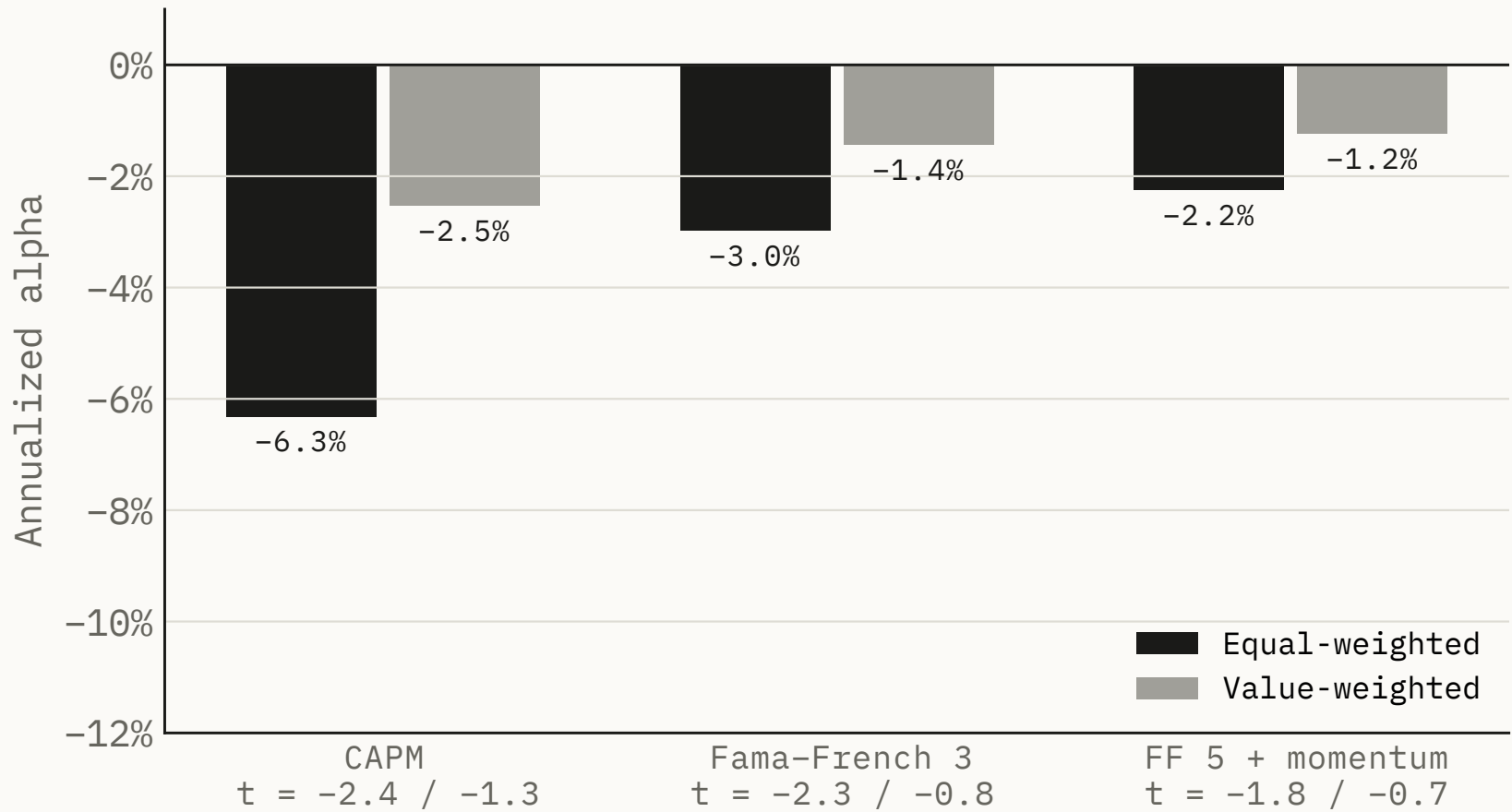
after 60 months · 641 deals with a five-year window

−18% to −11%

36-month mean across 9 specifications

# Most of the gap is the acquirer. Paying in stock is not.

ANNUALISED ALPHA BY MODEL · T-STATISTICS EQUAL / VALUE-WEIGHTED



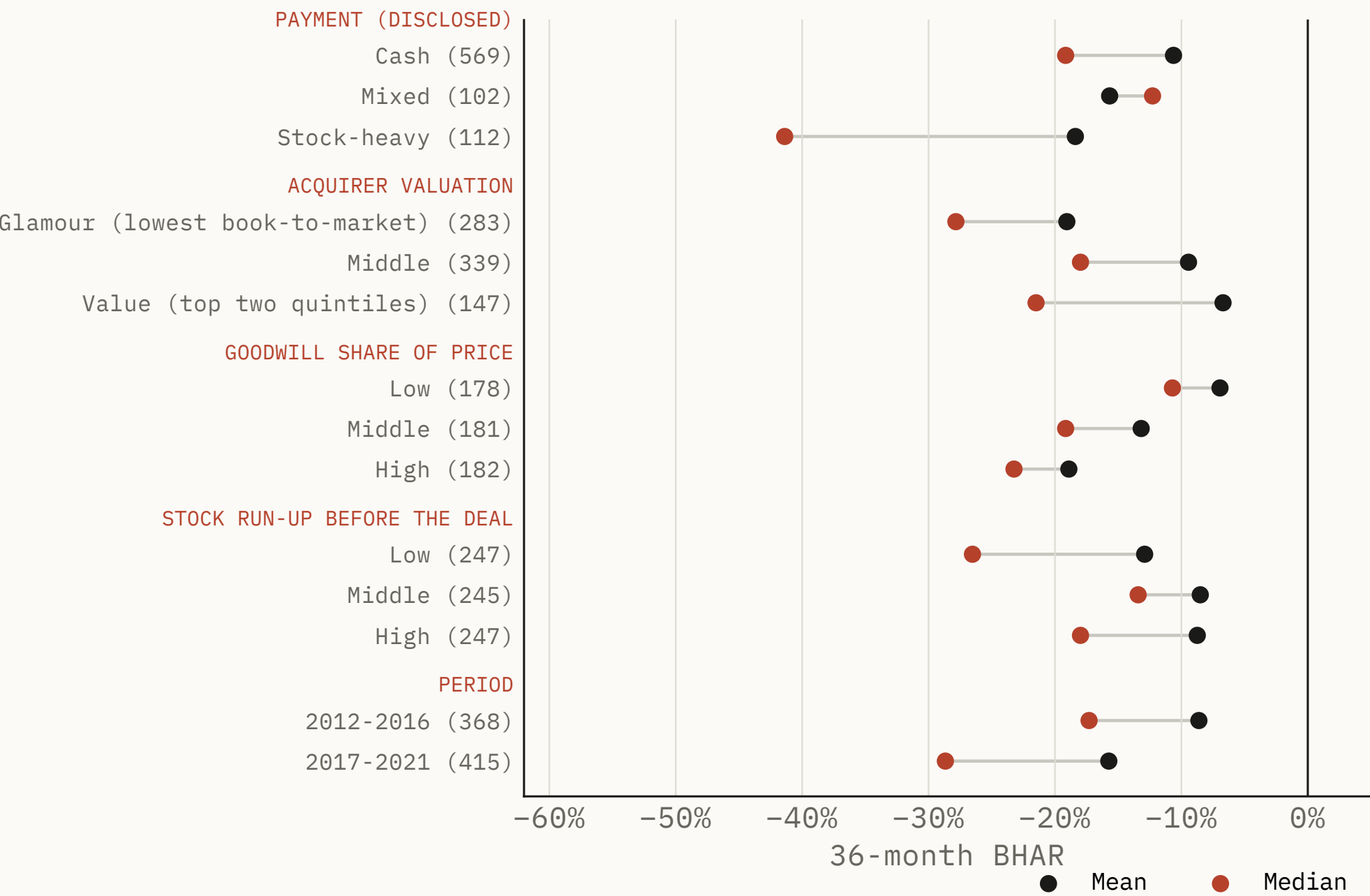
FIVE FACTORS + MOMENTUM, ALPHA PER MONTH

| PORTFOLIO                      | ALPHA  | T     | SIGNIF. |
|--------------------------------|--------|-------|---------|
| All acquirers, equal-weighted  | -0.19% | -1.80 | 10%     |
| All acquirers, value-weighted  | -0.10% | -0.72 | n.s.    |
| Paid in cash                   | -0.10% | -0.82 | n.s.    |
| Paid partly or mostly in stock | -0.58% | -3.18 | 1%      |
| Stock minus cash               | -0.54% | -2.65 | 1%      |

Loadings explain the gap: beta 1.14, small-cap tilt 0.78, momentum -0.13. Newey-West t-statistics, 153 months.

# Stock, glamour and goodwill point the wrong way.

36-MONTH BHAR · BLACK MEAN, RED MEDIAN · DEALS IN BRACKETS



-41.4%

median, paid mostly in stock · cash -19.2%

-19.1%

mean, glamour acquirers · value acquirers -6.7%

-18.9%

mean, highest goodwill share of price · lowest -6.9%

In one regression with all traits and industry, none is significant on its own ( $R^2$  0.04): warning signs, not a formula.

# No synergies in the margins; goodwill write-downs are common.

INDUSTRY-ADJUSTED OPERATING ROA, MEDIAN • FISCAL YEARS VS THE DEAL



**+3.8% → +2.1%**

adjusted ROA, year -1 to years +1 to +3

**-0.44%**

abnormal change after mean reversion,  $t = -1.14$

SHARE OF ACQUIRERS WITH A GOODWILL IMPAIRMENT



**47%**

impaired goodwill within five years

**33%**

impaired at least 10% of the deal value

# Four questions before approving a large deal.

## Start from the base rate

63% of acquirers trailed comparable stocks over three years. A deal case should say why this one is different, in terms that can be checked.

## Price the currency

Stock-funded acquirers lost about 7% a year after risk adjustment, cash acquirers 1.2%. Ask what the side taking shares believes about their value.

## Track the synergies

Margins did not improve beyond mean reversion. Tie the integration plan to margin and cash-flow milestones for years one to three.

## Stress the goodwill

Deals with the most goodwill did worst, and 47% of acquirers impaired within five years. Most of the price should be explained by the plan.

Limits: deals from XBRL spending and 8-K Item 2.01 dates rather than a deal database; payment relies on filer tags; 83% of candidates matched to prices; benchmarks do not control for industry; returns show association, not causation. Deliverables: interactive page, Colab notebook, Excel model with live formulas, paper, memo.